Optimal Spectral Allocation in Citizens Broadband Radio Service

Zahra Hoobakht, Harsha Gangammanavar, and Dinesh Rajan, *Senior Member, IEEE*

***Abstract*—This paper introduces a deterministic optimization framework for spectrum resource allocation, primarily focusing on the Citizens Broadband Radio Service (CBRS). The proposed framework aims to maximize spectrum utilization with minimum possible transmit power while ensuring that operational constraints are met for diverse users. We present mixed-integer linear programming (MILP) formulations for different spectrum allocation configurations, including contiguous and non-contiguous channels and uniform and non-uniform power allocation. To overcome the computational challenges encountered in solving MILPs for large-scale networks, we propose a computationally efficient heuristic method called Sequential Resource Allocation for Warm Start (STRAWS) that addresses resource allocation on one channel at a time. Using extensive computational experiments, we establish that, compared to the conventional contiguous model, the non-contiguous, non-uniform configuration demonstrates an average 15% improvement in low-demand and 75% in high-demand scenarios across all desired Signal-to-Noise ratios. The experiments also reveal that leveraging the STRAWS solution to warm start the optimization process enhances the overall solution quality within tight computational time limits. We also quantify the superior performance of STRAWS in several situations involving trade-offs between the number of users and the total network power. Despite a CBRS focus, our approach readily extends to cognitive radio, IoT, and vehicular networks.**

***Index Terms*—Citizen broadband radio service (CBRS), spectrum sharing, mixed-integer linear programming, cognitive radios, signal-to-noise ratio (SNR), non-contiguous frequency bands, non-uniform power allocation, real-time spectrum allocation.**

I. Introduction

FROM smartphones and smart appliances to autonomous vehicles and the Internet of Things (IoT), our society relies heavily on wireless networks to stay connected [1]. This growing reliance on wireless technology has exposed a critical challenge which is the scarcity of available bandwidth [2], [3]. Spectrum sharing is a wireless communication strategy that allows multiple users, services, or systems to share the same portion of the radio frequency spectrum simultaneously and efficiently and is a promising approach to address the spectrum scarcity [4], [5]. Unlike traditional fixed allocation, dynamic spectrum sharing employs spectrum database and tiered access models to effectively organize and coordinate spectrum access in response to the specific demands of users, geographical location, and temporal considerations [6], [7].

The Citizens Broadband Radio Service (CBRS) represents a significant advancement in the spectrum-sharing landscape [8]. In the United States, it operates within the 3.5 GHz band [9]. As shown in Fig. 1, CBRS operates on a three-tiered access system, consisting of Incumbent Access, Priority Access License (PAL), and General Authorized Access (GAA). Incumbent Access is reserved for critical users such as government and military entities [10]. PAL access is allocated through licenses, typically to commercial entities, while GAA allows unlicensed users to access available spectrum [11]. While this tiered structure serves as the foundational framework for spectrum allocation, a dynamic intelligent agent, known as the Spectrum Access System (SAS), resides at CBRS' core. SAS assumes a central role within the CBRS network, taking responsibility for real-time spectrum management and coordination [12], [13]. Based on channel requests from devices, SAS uses cloud-based servers and databases to identify available frequencies in specific geographic areas and then assigns them to users considering their priority, location, and the interference they may cause with the existing users [7], [14].

This paper introduces a deterministic optimization framework that addresses resource coordination among GAA users within the CBRS network. In our model, SAS intelligently assesses channel requests and can partially or fully approve these requests.

In this paper, the Mixed-Integer Linear Programming (MILP) approach, combined with the sequential resource allocation for warm-start (STRAWS) technique, has been proposed as a promising solution for maximizing spectrum utilization and ensuring allowable interference threshold in near real-time applications. To this aim, key contributions of this work are outlined as follows:

- We introduce a MILP framework that allows for the optimal allocation of channels to different transmitter-receiver (TxRx) pairs. This framework offers two distinctive features:
  1) *Non-contiguous band allocation* is employed for spectrum sharing, enabling the efficient utilization of fragmented or underutilized spectrum resources. Our

Manuscript received 2 November 2023; revised 22 February 2024; accepted 23 March 2024. Date of publication 3 April 2024; date of current version 7 June 2024. The associate editor coordinating the review of this article and approving it for publication was N. Zorba. *(Corresponding author: Zahra Hoobakht.)*

Zahra Hoobakht and Dinesh Rajan are with the Department of Electrical and Computer Engineering, Southern Methodist University, Dallas, TX 75275 USA (e-mail: hhoobakht@smu.edu; rajand@smu.edu).

Harsha Gangammanavar is with the Department of Operations Research and Engineering Management, Southern Methodist University, Dallas, TX 75275 USA (e-mail: harsha@smu.edu).

Digital Object Identifier 10.1109/TCCN.2024.3384491

2332-7731 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.

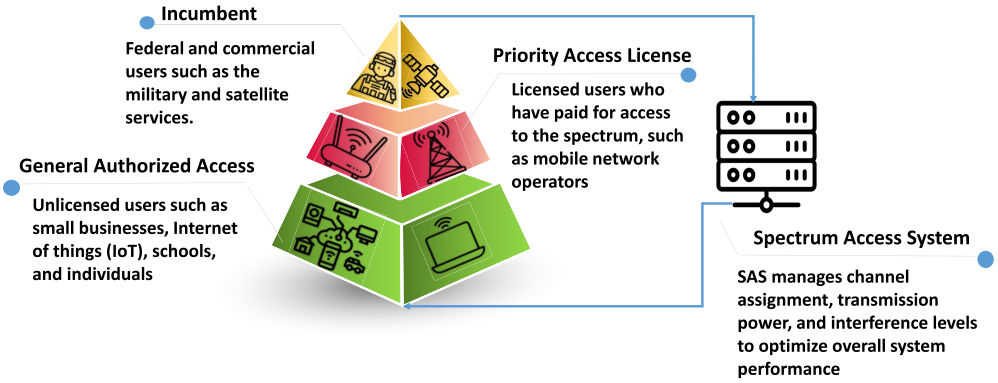


Fig. 1. CBRS hierarchical tiers including incumbent users, Priority Access License (PAL) holders, and General Authorized Access (GAA) users.

results demonstrate an average 15% improvement in low-demand scenarios and up to 75% in high-demand cases across all SNRs when employing non-contiguous channel allocation compared to contiguous channel allocation.

2) *Non-uniform power distribution* is employed to quantify the benefits of varying power across different channels. The results indicate that in low-traffic scenarios, non-uniform power allocation reduces power consumption by 25% than uniform power allocations.

- We develop a sequential channel allocation (STRAWS) method, in which one channel at a time is allocated to various spatially distributed users. After completing all channel allocations in this manner, the resulting allocation is feasible for the original optimization problems and is used as a warm start. This method is shown to be effective especially in low SNR cases, where the optimality gap of traditional solution to the MILP is high. We show that this method enhances the objective function by up to 50% in low-demand situations and by up to 20% in high-demand conditions.
- Our framework allows SAS to select the operating point that effectively balances the allocation of available channels with the total network transmit power. We quantify this trade-off and show that, for instance, at minimum desired SNR of 16 dB, a 15% increase in the number of allocated channels requires a corresponding 76% increase in transmit power.

The rest of the paper is organized as follows. In Section II, we discuss related work. Section III, presents the MILP model for channel and power allocation to CBRS users. In Section IV, the STRAWS method is presented. Numerical results as shown in Section V, and finally, conclusions are presented in Section VI.

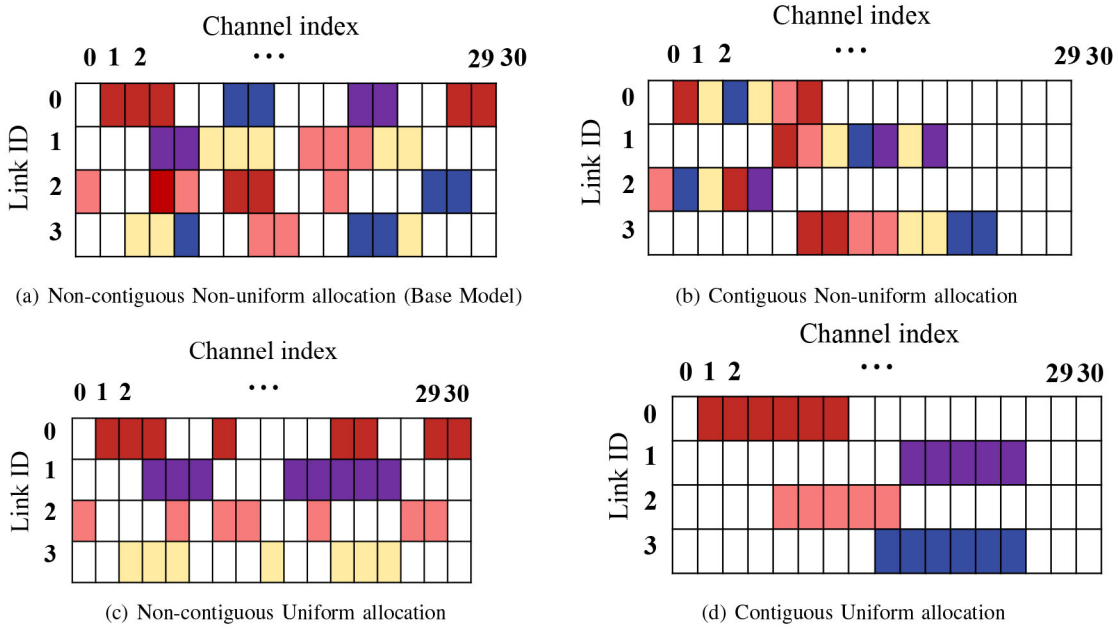
II. Related Work

The efficient use of the available spectrum is a complex problem that necessitates the optimization of several parameters, including frequency channel selection, power transmission, signal-to-noise ratio (SNR) level, and interference management.

Optimization challenges in spectrum sharing have been addressed and resolved through the employment of learning-based methods, heuristic algorithms, and deterministic frameworks. For instance, in [15], the authors introduced an auction algorithm, utilizing Q-learning for dynamic spectrum access. Multiple GAA users dynamically and intelligently bid using Q-learning to access PAL reserved idle channels, and channels are allocated based on maximizing their bidding offers. An auction-based approach for dynamic spectrum access that serves both on-demand and proactive traffic is proposed in [16]. In [11], a reinforcement learning-aided algorithm is proposed for use-or-share policies in the CBRS band sharing. This algorithm allows lower-priority GAA nodes to access unused higher-priority PAL spectrum. The authors in [17] address power allocation in a Non-Orthogonal Multiple Access (NOMA) system with multiple antennas facing a smart jammer. The problem is formulated as a zero-sum game between a base station and the jammer. A reinforcement learning-based power control scheme is proposed for downlink NOMA transmission, without knowledge of jamming and channel parameters. The use of Dyna architecture and hotbooting technique accelerates learning speed in Q-learning-based power allocation, enhancing NOMA transmission efficiency in the presence of smart jammers. Simulation results demonstrate a significant increase in user data rates and utilities compared to standard Q-learning-based strategies. In [18], the authors proposed a dedicated graph coloring algorithm for spectrum allocation, which allocates power and frequency resources among CBSDs under SAS control.

Although heuristic algorithms offer simplicity and speed, they may not always guarantee optimal CBRS channel allocation. There are limited studies using deterministic optimization techniques to maximize channel allocation and address interference problems by systematically finding optimal solutions through mathematical optimization techniques. In [19], optimal spectrum-sharing for a space information network comprised of a geostationary orbit satellite and a pair of low earth orbit satellites is modeled by a two-dimensional line search method which is an optimization technique used for highly nonconvex problems. In the application of autonomous driving, vehicular networks are used to provide energy-efficient spectrum sharing. A joint optimization problem is formulated in [20] as a mixed-integer nonlinear programming (MINLP) problem, which is solved by combining Hungarian and Dueling Deep Q Networks (DDQN). The proposed design

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.



(a) Non-contiguous Non-uniform allocation (Base Model)

(b) Contiguous Non-uniform allocation

(c) Non-contiguous Uniform allocation

(d) Contiguous Uniform allocation

Fig. 2. CBRS channel and power allocations in a) Non-contiguous Non-uniform (Base Model), b)Non-contiguous Uniform, c) Contiguous Non-uniform and d) Contiguous Uniform configurations. The colors represent power types, and the blocks represent channel allocation types. In Fig. 2(a), we can observe that in one scenario, 9 channels are allocated to Link 0 in a non-contiguous and non-uniform manner. In another situation, Fig. 2(b), 6 contiguous channels are allocated to Link 0.

framework in [21] presents a novel approach to synthesize the power pattern reflected by Reconfigurable Intelligent Surfaces (RIS) to meet customized requirements for broad coverage. Specifically tailored for the communication of broadcast channels, this framework extends the concept of broad coverage from a single transmit stream to scenarios involving multiple streams. In the context of a multi-user Orthogonal Frequency Division Multiplexing Access (OFDMA) system, the authors employ quasi-static broad coverage and derive analytical expressions for the downlink rate. Their results indicate a logarithmic increase in the downlink rate with the power gain reflected by the RIS, suggesting the potential efficacy of RIS in enhancing communication performance.

In [22], [23], the authors implemented game theory and reinforcement learning within SAS to reduce interference for incumbents and PAL devices in spectrum sharing. In [24], a permissioned blockchain trust framework is proposed for enabling effective spectrum sharing in wireless communication networks. A smart contract is designed on the constructed blockchain to facilitate spectrum trading among multiple users, thereby demonstrating privacy and fairness [25].

While the aforementioned studies offer valuable insights, there is still potential for enhancing spectrum and power utilization. This paper aims to demonstrate that improvements in spectrum utilization can be achieved through the implementation of non-contiguous channel allocation and non-uniform power distribution. Additionally, by employing the STRAWS method, we can effectively narrow the optimality gap and provide near-optimal solutions for deterministic optimization in large-case scenarios.

III. Model Formulations

In this section, we present an MILP-based approach to tackle the challenge of optimizing resource allocation in CBRS networks. We begin by outlining the study setting and introduce our proposed resource allocation optimization models.

A. System Setting

Consider a set $\mathcal{D}$ of CBRS devices (CBSDs), indexed by $i \in \mathcal{D}$. We assume that these devices are stationary and randomly located at Cartesian coordinates (x_i, y_i); an example of CBSD placement is shown in Fig. 3. The distance between devices i and j is denoted by d_{ij}. Each CBSD is capable of communicating over a set $\mathcal{N}$ of channels within the spectrum. We define set $\mathcal{L}$ to represent the communication requests. Each communication request, denoted as $(i, j) \in \mathcal{L}$, is characterized by a transmitter device $i \in \mathcal{D}$ that requests the allocation of b_{ij} channels to communicate with receiver device $j \in \mathcal{D}$. The SAS will allocate as many channels as possible to each of the requests subject to the following conditions being met: 1) The interference level at each operating node remains below a specified threshold P_{int}. 2) The received signal power is higher than the sensing threshold of the receiver, denoted as P_{sense}. 3) The total transmitted power of a radio must not exceed its maximum power limit, specified as $\bar{P}_{\text{i}}$ for all $i \in \mathcal{D}$. 4) The transmitted power on every channel must not exceed the regulatory limit denoted as $\underline{P}_{\text{n}}$ for all $n \in \mathcal{N}$. 5) Received signal level must ensure the desired quality of service in CBRS networks. We quantify the quality of service by a minimum SNR threshold represented by δ.

B. Optimization Model for the Base Configuration

We consider a general channel allocation method in which each requests $(i, j) \in \mathcal{L}$ can be allocated a set of non-contiguous bands. Further, the transmit power on each band can also be varied. An example of such an allocation is shown in Fig. 2(a). The binary variable z_{ijn} serves as the

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.

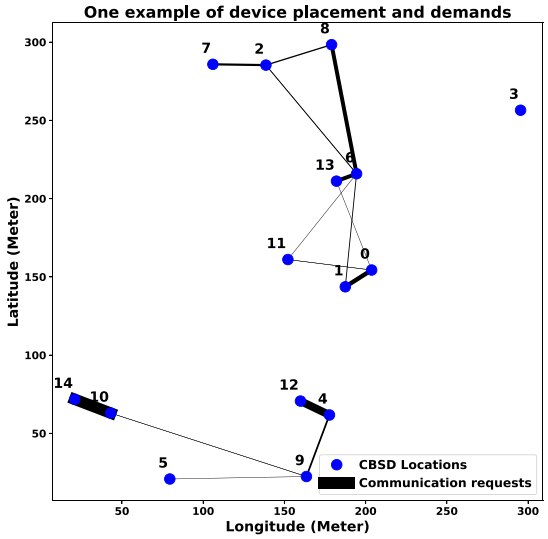


Fig. 3. An example of the CBRS device locations and communication requests. The thickness of lines is proportional to the number of channel requests.

channel allocation indicator, where $z_{ijn} = 1$ indicates that channel $n \in \mathcal{N}$ is allocated to the communication link (*i*, *j*), while $z_{ijn} = 0$ indicates no channel allocation. The continuous variable p_{in} represents the transmit power of user *i* on channel *n*. The variable q_{ijn} represents the received signal power from transmitter *i* at receiver *j* on channel *n*.

The following constraint ensures that the total number of allocated channels to each requested link does not exceed the demand:

$$\sum_{n\in\mathcal{N}} z_{ijn} \le b_{ij} \quad \forall (i,j) \in \mathcal{L}. \tag{1}$$

The total transmission power of a radio must not exceed the maximum power limit, a constraint which is ensured as:

$$\sum_{n\in\mathcal{N}} p_{in} \le \bar{P}_{\mathrm{i}} \quad \forall i \in \mathcal{D}. \tag{2}$$

While the above constraints link the resource allocation across all the channels, the subsequent constraints apply to a specific channel *n* in $\mathcal{N}$. At a particular device *i*, we ensure that a specific channel *n* is allocated to at most one communication request as:

$$\sum_{j\,:\,(i,j)\in\mathcal{L}} z_{ijn} \le 1 \quad \forall i \in \mathcal{D}. \tag{3}$$

Notice that if a channel is allocated to a communication link, then the transmission power must be non-negative and must not exceed the maximum regulatory allowable power for each channel. We guarantee this using the following constraint:

$$0 \le p_{in} \le \underline{P}_{\mathrm{n}} \left(\sum_{j:(i,j)\in\mathcal{L}} z_{ijn} \right) \quad \forall i \in \mathcal{D}. \tag{4}$$

If channel *n* is not allocated to any of the requests from node *i*, then the upper bound of the above constraint equals zero forcing, $p_{in} = 0$.

Now, we discuss the received power in CBRS networks. The Friis propagation model is used to analyze the behavior of electromagnetic waves as they propagate through space [26]. The received power q_{ijn} is given by:

$$q_{ijn} = \alpha d_{ij}^{-\gamma_j} p_{in} \quad \forall i,j \in \mathcal{D}, \tag{5}$$

where γ is the path-loss exponent, which characterizes how the received signal power decreases with distance depending on the propagation environment, and α captures the combined effect of the various factors such as antenna gains and wavelength.

Any receiver is capable of detecting signals with power levels higher than the defined sensitivity threshold. Therefore, for successful communication the received power should satisfy:

$$q_{ijn} \ge P_{\mathrm{sense}} z_{ijn} \quad \forall (i,j) \in \mathcal{L}. \tag{6}$$

When channel *n* is assigned to a communication link (*i*, *j*), i.e., $z_{ijn} = 1$, we must ensure that the interference caused by other devices $k \neq i$ is within the tolerable limits. We capture this requirement by:

$$\sum_{\substack{k\in\mathcal{D}\\k\neq i}} q_{kjn} \le P_{\mathrm{int}} + \big(1 - z_{ijn}\big) M \quad \forall (i,j) \in \mathcal{L}, \tag{7}$$

where $M > 0$ is a scalar (referred to as 'big-M' in optimization literature). The parameter *M* is selected sufficiently large such that when $z_{ijn} = 0$ constraint (7) is automatically satisfied for all $(i,j) \in \mathcal{L}$. Notice that the above constraint captures co-channel interference at receiver *j* for every communication request that is granted. Interference from communication requests allotted to adjacent channels can be captured by including the received power for these requests. However, we assume that adjacent channel interference is negligible and therefore ignore it from the model. According to [27] the effect of adjacent channel interference can be controlled by implementing bandpass filters in the receiver. When $z_{ijn} = 1$, we must also guarantee that a desired minimum SNR is met for each link. We accomplish this SNR requirement by the following:

$$q_{ijn} - \delta \left(\sum_{\substack{k\in\mathcal{D}\\k\neq i}} q_{kjn} + \sigma^2 \right) + M\big(1 - z_{ijn}\big) \ge 0 \quad \forall i,j \in \mathcal{D}. \tag{8}$$

where σ^2 is effective noise power.

The objective function, $f(z,p)$, of the MILP model is given by:

$$f(z,p) = \sum_{n\in\mathcal{N}} \underbrace{\left[\sum_{(i,j)\in\mathcal{L}} z_{ijn} - \lambda \sum_{i\in\mathcal{D}} p_{in} \right]}_{f_n(z_n,p_n)},$$

where $z_n = [z_{ijn}]_{\forall (i,j)\in\mathcal{L}}$ and $p_n = [p_{in}]_{\forall i\in\mathcal{D}}$ are channel-specific decision vectors, and f_n is a channel specific objective function. Further, $z = [z_n]_{\forall n\in\mathcal{N}}$ and $p = [p_n]_{\forall n\in\mathcal{N}}$ are

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.

the consolidated decision vectors. The parameter λ controls the trade-off between maximizing the channel allocation and minimizing power consumption. A higher value of λ places more emphasis on reducing power consumption, which leads to more conservative power allocations. The optimization model for the base configuration is expressed as:

$$\begin{aligned}&\text{Maximize } f(z,p)\\&\text{Subject to: (1)}-\text{(2) \& (3)}-\text{(8)} \quad \forall n \in \mathcal{N}. \qquad \text{(P1)}\end{aligned}$$

Going forward, we refer to the above optimization problem as the base model.

C. Optimization Models for Other Configurations

In some cases, the CBSDs may be restricted to utilize contiguous channel blocks. To quantify the advantages of non-contiguous channels and non-uniform power allocation, we study a variant of the base model in which all the channels allocated to a request are in contiguous bands. An illustrative example can be seen in Fig. 2(b). Similarly, to understand and quantify the benefits of allocating different power levels to each band, we study another variation of the base model in which constant power is allocated to the possibly non-contiguous bands. An example is shown in Fig. 2(c). Finally, we also study the configuration where the bands are allocated contiguously, and the transmit power is constant across all allocated bands. The Fig. 2(d) depicts an example.

To model contiguous channel allocation, we introduce the set $\mathcal{A}$, representing the set of contiguous bandwidth allocation patterns. Given the total number of channels $|\mathcal{N}| = \mathcal{N}$, there are $|\mathcal{A}| = A = \frac{N(N+1)}{2}$ possible allocation patterns. These patterns are described by an $A \times N$ matrix Γ, where the matrix elements, denoted as γ_{an}, have a value of 1 if channel n is allocated in pattern a, and 0 if channel n is not allocated in that particular pattern a, for all $n \in \mathcal{N}$ and $a \in \mathcal{A}$. We use a binary variable y_{ija} to allocate the contiguous pattern a to the communication link (i,j). The contiguous channel allocation is modeled using the next two constraints that are defined for $\forall n \in \mathcal{N}$. To ensure that at most one contiguous pattern is allocated to the communication link (i,j), we include

$$\sum_{a \in \mathcal{A}} y_{ija} \leq 1 \quad \forall (i,j) \in \mathcal{L}. \tag{9}$$

Next, to ensure that channel n is allocated to link (i,j) ($z_{ijn} = 1$) if and only if it is part of the selected contiguous pattern a ($\gamma_{an} = 1$) and pattern a is chosen ($y_{ija} = 1$), we have

$$z_{ijn} = \sum_{a \in \mathcal{A}} \gamma_{an} y_{ija} \quad \forall (i,j) \in \mathcal{L}. \tag{10}$$

Using the above, the optimization model for contiguous channel and non-uniform power allocation is formulated as:

$$\begin{aligned}&\text{Maximize } f(z,p)\\&\text{Subject to: (1)} - \text{(2) \& (3)} - \text{(10)} \quad \forall n \in \mathcal{N}. \quad \text{(P2)}\end{aligned}$$

To model configurations that mandate uniform power allocations to all allocated channels, consider the following constraint:

$$p_{in} = \sum_{j:(i,j)\in\mathcal{L}} v_i z_{ijn} \quad \forall i \in \mathcal{D}, n \in \mathcal{N},$$

where the decision variable v_i represents uniform power allocated. Notice that the right-hand side of the above equation contains bilinear terms, each involving a continuous variable v_i and a binary variable z_{ijn}. To avoid the computational difficulties associated with handling nonlinear constraints, we employ McCormick envelopes to obtain an equivalent linear representation of the above constraints [28]. This linear representation is given for all $n \in \mathcal{N}$ by:

$$p_{in} = \sum_{(i,j)\in\mathcal{L}} w_{ijn}, \tag{11a}$$

$$w_{ijn} \geq 0 \quad \forall (i,j) \in \mathcal{L}, \tag{11b}$$

$$w_{ijn} \leq \pi_{in} z_{ijn} \quad \forall (i,j) \in \mathcal{L}, \tag{11c}$$

$$w_{ijn} \leq v_i \quad \forall (i,j) \in \mathcal{L}, \tag{11d}$$

$$w_{ijn} \geq v_i + \pi_{in} z_{ijn} - \pi_{in} \quad \forall (i,j) \in \mathcal{L}, \tag{11e}$$

where the $\pi_{\text{in}} = \min(\bar{P}_{\text{i}}, \underline{P}_{\text{n}})$. The optimization model for non-contiguous channels and uniform power allocation is formulated as follows:

$$\begin{aligned}&\text{Maximize } f(z,p)\\&\text{Subject to: (1)} - \text{(2) \& (3)} - \text{(8), (11)} \quad \forall n \in \mathcal{N}. \quad \text{(P3)}\end{aligned}$$

Finally, our contiguous channel and uniform power allocation model will be addressed as:

$$\begin{aligned}&\text{Maximize } f(z,p)\\&\text{Subject to: (1)} - \text{(2) \& (3)} - \text{(11)} \quad \forall n \in \mathcal{N}. \quad \text{(P4)}\end{aligned}$$

Notice that the optimization models for non-contiguous configurations, viz., (P1) and (P3), involve only two constraints (1) and (2) that couple the decision variables across all the channels. We exploit the structure of non-contiguous allocation to design a warm-starting procedure that we describe next. When contiguous channel allocation, as in (P2) and (P4), is involved, the allocation patterns due to the contiguous nature of the allocation introduce further interactions between channel-specific decision variables.

Notice that the decision variable that captures received power can be dropped from the models due to equation (5). In its compact form, (P1) has $N(D+L)$ variables and $L + D + 2ND + 2LN + D^2N$ constraints. This indicates that the problem scales linearly in the number of channels and quadratically in the number of devices. The model for contiguous nonuniform configuration (P2) has additional $LN(N+1)/2$ variables and $2L$ constraints. Therefore, this model scales quadratically even in the number of channels. Next, in relation to (P1), (P3) has additional $D + NL$ variables and $2ND + 4NL$ constraints to capture uniform power allocation and McCormick relaxation of the resulting bilinear terms. Finally, in (P4) relative to (P1), we have additional $D + NL + LN(N+$

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.

$1)/2$ variables and $2ND + 4NL + 2L$ constraints. While the computational complexity of the general MILPs is not settled, there is a theory to suggest that they are not polynomial [29]. Nevertheless, the above-noted dependence on crucial model parameters such as the number of devices and channels shows how the problem scales in size.

IV. Sequential Resource Allocation for Warm Start (STRAWS)

We propose the STRAWS method to solve the optimization problems for CBRS configurations involving non-contiguous channel allocation. The STRAWS method aims to identify a feasible solution to the problem computationally efficiently, which can be used to warm start the MILP solution algorithms.

Notice that in non-contiguous models (P1) and (P3) constraints (1) and (2) couple decisions across all the channels while the other constraints are channel specific. This characteristic motivates the design of STRAWS, where resources are allocated one channel at a time. Therefore, STRAWS involves N steps. We describe the steps of the algorithm in step $k \in \mathcal{N}$.

At the beginning of step k, we have allocated resources to the first $k-1$ channels. The optimal solutions $z^{k-1} = (z_1^\star, z_2^\star, \ldots, z_{k-1}^\star)$, $p^{k-1} = (p_1^\star, p_2^\star, \ldots, p_{k-1}^\star)$, and $q^{k-1} = (q_1^\star, q_2^\star, \ldots, q_{k-1}^\star)$ are known. Let $\mathcal{D}^k \subseteq \mathcal{D}$ denote the set of devices whose power capacity has not been completely utilized so far. For $i \in \mathcal{D}^k$, we denote by $\overline{P}_i^k$ the remaining power which is given by:

$$\overline{P}_i^k = \bar{P}_{\mathrm{i}} - \sum_{m=1}^{k-1} p_{im}^\star. \tag{12}$$

Similarly, we denote by $\mathcal{L}^k \subseteq \mathcal{L}$ a set of links (i, j) whose transmitter has enough power to transmit (i.e., $i \in \mathcal{D}^k$) and whose bandwidth requests have not been completely fulfilled. The remaining bandwidth request is given by:

$$\bar{b}_{ij}^k = b_{ij} - \sum_{m=1}^{k-1} z_{ijm}^\star. \tag{13}$$

Using the above, we state the channel-specific optimization problem as follows:

$$\begin{aligned} \text{Maximize} \quad & f_k(z_k, p_k) \\ \text{Subject to:} \quad & z_{ijk} \le \bar{b}_{ij}^k \quad \forall (i,j) \in \mathcal{L}^k \\ & p_{ik} \le \overline{p}_i^k \quad \forall i \in \mathcal{D}^k \\ & (3) - (8). \end{aligned} \tag{P5}$$

The above optimization problem is solved to obtain the optimal solution $(z_k^\star, p_k^\star, q_k^\star)$ which is appended to $(z^{k-1}, p^{k-1}, q^{k-1})$ to obtain (z^k, p^k, q^k). This completes one step of STRAWS.

We use the terminal solution obtained after N steps of STRAWS, i.e., (z^N, p^N, q^N), to warm start the MILP optimization algorithm to solve the extensive for of (P1). We expect that this method will significantly expedite the optimization of the base model by starting from a promising solution and potentially leading to faster convergence. Recall the number of variables and constraints in the base model scale linearly with the number of channels. STRAWS decomposes the problem by channels; where starting with $D + L$ variables and $3L + 3D + D^2$ constraints, the size of the subproblems gradually reduces over the iterations as we allocate channels to the communication links. This reduction in problem size helps STRAWS achieve its computational efficiency. Since individual subproblems are still MILPs and we only use the STRAWS solution to warm-start, the overall complexity of the analysis of this approach is untenable. Therefore, we verify the performance of STRAWS through numerical experiments that we report in the next section.

TABLE I
SAS Decision-Making Model Parameters

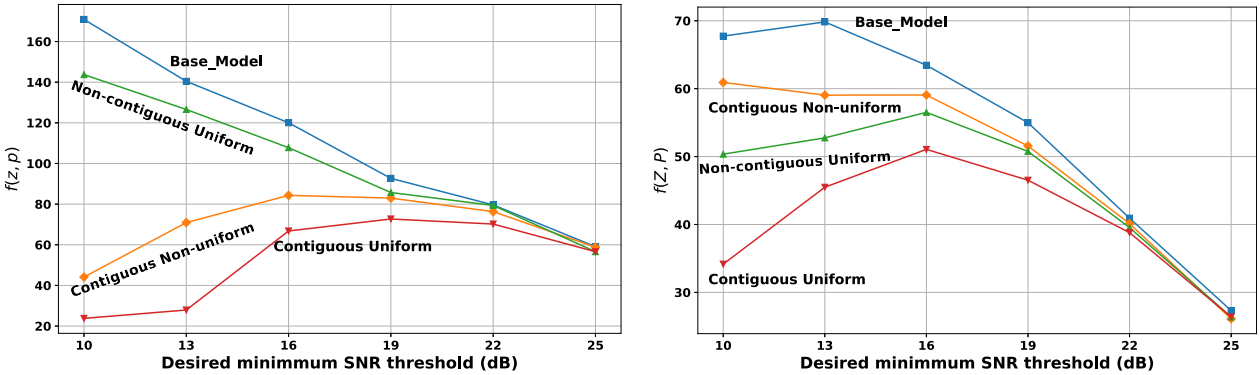
Parameter	Value
Maximum power of every channel ($\underline{P}_{\mathrm{n}}$)	0.2 W
Maximum power of every radio ($\bar{P}_{\mathrm{i}}$)	2 W
P_{int}	10^{-6} W
P_{sens}	10^{-8} W
Noise power (σ^2)	10^{-6} W
Weighting factor λ	[0, 0.1, 1, 5, 10]
Desired SNR values δ	[10, 13, 16, 19, 22, 25] dB
Path-loss exponent (γ)	range [2,3]
Free space constant (α)	1

V. Experimental Evaluation

In this section, we evaluate the performance of different MILP models proposed in Section III using extensive numerical experiments. In our experiments, we generate sets of $|\mathcal{D}| = 30$ transceivers that are randomly distributed within a 300×300 m^2 area. The entire spectrum between 3550 to 3700 MHz is divided into $N = 30$ channels, each with 5 MHz bandwidth. The static parameters used in the optimization models are listed in Table I. These parameter values have been chosen to comply with regulations and align with CBRS device standards as outlined in [30] and [31]. Note that the choice of $\gamma = 2$ and $\alpha = 1$ implies free-space propagation without any obstacles. We investigate two different demand settings to assess the SAS decision-making process. The "high-demand" condition simulates peak usage, with each user requesting nearly all available bandwidth. On the other hand, the "low-demand" condition represents the scenarios in which users request up to a maximum of five channels due to their specific application.

We performed 30 replications of each experiment by randomly and independently generating transceiver coordinates and link bandwidth requests. For a given realization of coordinates and requests, we generate optimization problem instances for all combinations of interest using the static

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.



(a) Average Resource Allocation Across 30 Realizations in High-demand (b) Average Resource Allocation Across 30 replications in Low-demand

Fig. 4. Resource Allocation for different demand conditions across 30 replications for path-loss exponent $\gamma = 2$ and weighting factor $\lambda = 1$.

parameters in Table I. We used a MILP solver to directly solve the problem instances in the extensive form, and the STRAWS method to solve the base model (P1). In either case, we run the optimization process for a maximum of 10 minutes. The terminal solution is feasible for the optimization problem, but it is not guaranteed to be optimum. When the solution is not optimum, we monitor the optimality gap and the associated objective function value. While reporting our results, we use objective function values and solutions averaged across all 30 replications.

We conducted all our experiments using Python, version 3.9.7. We used the Python interface of Gurobi, version 9.5.1, and its MILP solver to build and solve the optimization models for all the resource allocation configurations outlined in Section III, respectively. All our experiments were conducted on the High-Performance Computing cluster, Maneframe II, which boasts a computational capability of 870 TFLOPS. The cluster consists of 354 nodes, each equipped with Intel CPU cores (AVX2) for a total of 11,276 cores. Additionally, Maneframe II is equipped with 275,968 accelerator cores and has a substantial total memory capacity of 120 TB (122,880 GB). For more detailed information, please refer to.[1]

A. Examine the Most Effective Resource Allocation Configuration Across the Desired Range of SNRs

We first examine the performance of CBRS spectrum allocation configurations at different SNR levels and demand conditions. This experiment is conducted for SNR values [10, 13, 16, 19, 22, 25] in dB and all four configurations. The results are presented in Figures 4(a) and 4(b) for high-demand and low-demand, respectively. The horizontal axis represents SNR threshold values, while the vertical axis represents the objective function values for different configurations.

In our models, lowering the SNR threshold values makes the constraint (8) less restrictive, allowing more users to become eligible for channel allocation. Therefore, we might expect the objective function value to decrease with an increase in the SNR threshold for each configuration. However, we observed that the optimization problems become progressively more difficult computationally with the increase in number of potential users. Consequently, at low SNR threshold values, the optimization process was terminated at the end of the allotted time frame with larger gaps than at higher SNR threshold values. For instance, for the non-contiguous, non-uniform configuration, the terminal optimality gap was 423% at $\delta = 10$ dB, while the problem was solved to optimality (i.e., 0% optimality gap) at $\delta = 25$ dB. Note that the solver reports the best feasible solution identified in the allotted time when the optimization process is terminated with a non-zero optimality gap. For any configuration, the decreasing trend in objective function value is observed only at higher SNR threshold values: $\delta \geq 16$ dB for low-demand conditions and $\delta \geq 19$ dB for high-demand conditions. Irrespective of whether the optimization process terminated with the optimal or the best feasible solution, we observe that our base model consistently outperforms all other configurations across all SNR threshold values and for both demand conditions. The ability to allocate non-contiguous channels and non-uniform power offers flexibility in the optimization model. The base model's performance is attributed to the presence of both these features.

In Fig. 4(a), the results for all four configurations under the high-demand condition are illustrated. We observe that our non-contiguous channel configurations have better performance when compared to their contiguous configuration counterparts. This behavior is attributed to the flexibility offered by non-contiguous channel allocation. Particularly, non-contiguous allocation can split the demand into smaller blocks and allocate them to channels with less interference, thus improving the overall spectrum resource utilization. On the other hand, in contiguous allocation, we identify the longest block that can be supported using an allocation pattern, and the remaining unmet demand is discarded. This allocation strategy negatively impacts utilization and is observed more prominently in high-demand conditions.

We observe that in low-demand conditions (see Fig. 4(b)), configurations with non-uniform power allocation have better performance compared to the uniform allocation methods. This advantage arises from the non-uniform power allocation's ability to allocate variable power levels to users, particularly those at a close distance to the transmitter. In other words, it conserves energy from users with lower power requirements

[1]https://www.smu.edu/provost/crc

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.

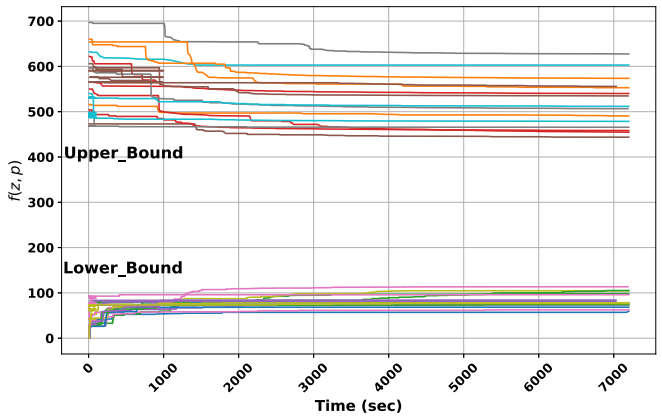


Fig. 5. Temporal behavior of upper and low for the base model with SNR threshold $\delta = 10\,\text{dB}$, path-loss exponent $\gamma = 2$ and weighting factor $\lambda = 1$.

and allocates that power to distant users. Furthermore, such a power allocation strategy also reduces interference, thereby, increasing the number of users who can be supported. The advantage offered by non-contiguous channel allocation is less significant when the demand is low (i.e., the need for long blocks of channels is smaller). In this case, the effect of non-uniform power allocation is more dominant.

Finally, comparing Figures 4(a) and 4(b), we observe that, for a given configuration and SNR threshold value, the objective function value is higher in high-demand conditions than in low-demand conditions. This behavior is due to the fact that the number of requests for the two demand conditions is similar, while the users request more channels in the high-demand compared to the low-demand conditions. These experiments lead us to conclude that, the base model, which captures non-contiguous channel allocation and non-uniform power allocation, has superior performance and must be considered in CBRS resource allocation. Therefore, we confine the rest of the experiments to the base model.

B. Solver Performance Enhancement Through STRAWS

As mentioned earlier, optimization models for low-SNR threshold values are computationally difficult. To assess the computation behavior, we ran experiments with the base model with low SNR threshold values for a longer (two hours) runtime restriction. Figure 5 shows the upper and the lower bound identified by the solver over the entire runtime for each of the 30 replications. While we observe significant improvements in lower and upper bounds during a run, the gap between these bounds remains high even after two hours. This gap indicates that either the solver is unable to identify a good solution or it is unable to certify optimality. In either case, we conclude the terminal solution is the best feasible solution. These results demonstrate the computation difficulty with this class of optimization problems.

Ensuring consistently high SNR levels for all users in a network is not always practical in real-world network deployments. Some users inevitably operate with lower SNR thresholds because achieving a high SNR often necessitates allocating higher power levels to individual users. To address the computational difficulty of low-SNR models, we proposed the STRAWS algorithm to identify the initial feasible solution in Section IV along with its associated objective function (P5). To compare the performance of directly invoking the optimization solver and warm-starting the optimization with STRAWS solutions, we conducted additional experiments. In these experiments, we solve 30 replications of the base model for a runtime of 10 minutes. When the STRAWS is used for warm-starting, the 10-minute restriction is imposed on the total time spent in executing both STRAWS and optimization. The results are shown in Fig. 6.

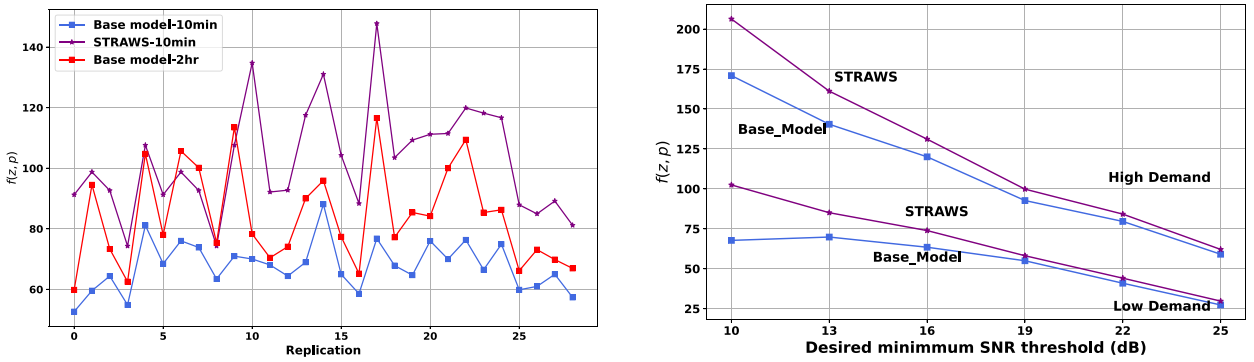
Fig. 6(a) shows the objective function value associated with the terminal solution for all replications. In addition to the 10-minute runtime experiments, the figure also includes the results from running the optimization process for the runtime of two hours. Comparing the base model results for 10 minutes and two hours, we observe an improvement in objective function value when the optimization process is run for a longer time. We also observe a significant improvement in objective function value when the optimization process with a 10-minute runtime is warm-started using the STRAWS solution. This improvement ranges between 17% observed in replication 9 to 92% in replication 18. The average improvement across all 30 experiments that STRAWS method achieves is 52%.

In Fig. 6(b), we present the average objective function value in the base model with and without warm-starting using the STRAWS algorithm. The results are for different SNR threshold values and two demand conditions. We observe that we attain better objective function values when warm-starting is used in all cases. For instance, when SNR is 10 dB, the results indicate that in the low-demand scenario, the STRAWS approach yields a 50% objective function value improvement over the base model. In the high-demand scenario, this advantage is 20%. Based on these experiments, we conclude that utilizing a computationally efficient STRAWS algorithm to identify an initial solution for warm starting has a significant advantage for optimizing non-contiguous CBRS resource allocation problems.

C. Trade-Off Between Channel Allocation and Power Distribution

There is an inherent trade-off between the total network transmit power and the number of channels allocated in the CBRS context. On the one hand, the allocation of sufficient transmit power is essential to ensure effective signal transmission, overcoming signal attenuation and the inherent limitations of radio propagation. On the other hand, higher transmit power levels increase the interference with neighboring links, potentially constraining the SAS from allocating additional channels. To explore the dynamics of this trade-off, we varied the weighting factor λ in the objective function (9) of the base model over the set $[0, 1, 5, 10]$. By increasing λ, we prioritize power reduction, resulting in lower transmit power levels and potentially sacrificing some channel allocation. Therefore, a high value of λ is suitable for scenarios emphasizing energy conservation. Conversely, decreasing λ incurs higher transmit powers, favoring channel allocation. This scenario is ideal for situations where spectral efficiency

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.



(a) performance comparison of base model after 10 minutes and 2hr with sequential in 10 minutes in desired SNR $\delta = 10$ dB in low demand condition

(b) Performance analysis between base model and STRAWS method at low and high demand

Fig. 6. Resource Allocation for different demand conditions across 30 replications for path-loss exponent $\gamma = 2$ and weighting factor $\lambda = 1$.

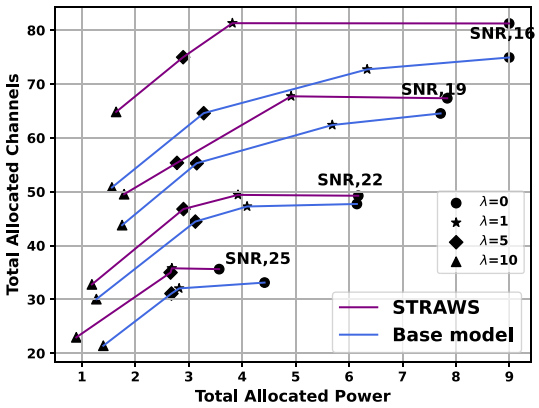


Fig. 7. Trade off between channel and power allocations for different weighting factor $\lambda = [0, 1, 5, 10]$ and for path-loss exponent $\gamma = 2$.

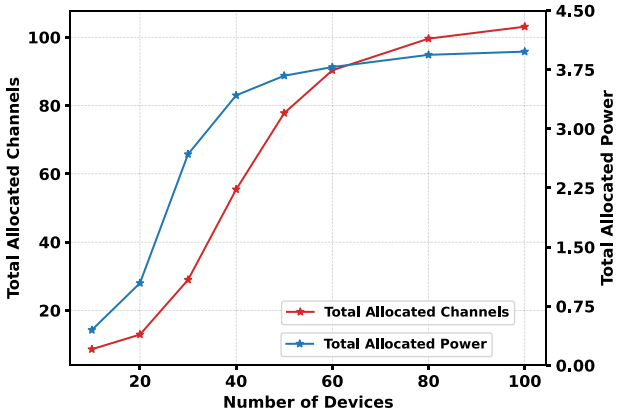


Fig. 8. Impact of device increment on channel and power allocation at High SNR level ($\delta = 25$ dB) using STRAWS allocation method in low-demand condition.

is preferred. The results, obtained with and without STRAWS, of the experiments conducted with different SNR threshold values, are shown in Fig. 7. The horizontal and vertical axes capture, respectively, the total power and channels allocated. The markers on each curve correspond to distinct λ values. These results show the continued benefits of STRAWS. It is clear from the figure that increasing the power beyond a certain value only results in marginal increases in the number of allocated channels and there are diminishing returns with further increases in the power. For example, the results for 16 dB obtained by using the STRAWS algorithm show that by increasing the power allocation from 1.64 to 2.89 (76%), the number of averaged channels allocated increased from 64 to 75 (17%). On the other hand, when power allocation increased from 3.6 to 8.9, the number of additional channels increased only by 2.5%. In other words, a 15% increase in the number of channels requires a corresponding 76% increase in power. This latter observation provides valuable insights for making informed decisions in the CBRS setting, particularly for capturing the decision-makers' preferences regarding increasing the revenue (realized by an increased number of channel allocation) and reducing power utilization.

D. Impact of Device Increment on STRAWS Resource Allocation

In this section, we investigate the influence of device increment on our advanced proposed method, STRAWS resource allocation, under high desired SNR ($\delta = 25$ dB) and in low-demand conditions. The rationale behind selecting this specific SNR threshold and demand condition lies in their combined effect of tightening constraints. A high desired SNR applies tighter constraints, while low-demand conditions result in a reduced number of requests for each user, effectively narrowing the decision-making space in the optimization process These deliberate choices are aimed at minimizing optimization time and narrowing the gap, thereby enhancing the reliability and adding certainty to the obtained results. The results, presented in Fig. 8, show that as the number of devices increases, both the number of channel and power allocations also increase. However, it can be seen that while the overall trend of allocation demonstrates an increase, the rate of increase slows down. The increasing interference between links, correlated with the rising number of devices, is the primary factor contributing to this slowdown. As a result, the

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.

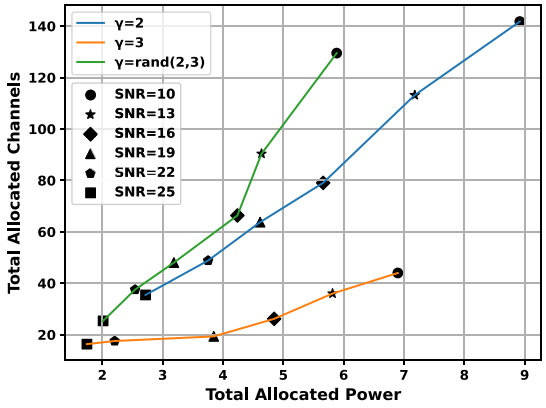


Fig. 9. Impact of changing loss exponent (γ) on resource allocation using STRAWS method at different SNR levels in low-demand condition.

SAS is unable to allocate additional resources to users to prevent interference among them.

E. Examine the Impact of Changing Loss Component (γ) on Resource Allocation

As the final experiment, to validate the real-world applicability of the STRAWS method for dynamic resource allocation in communication networks, we conduct an experiment investigating its robustness and adaptability under diverse propagation environments. We randomly vary the path loss exponent (γ) within the interval of [2, 3] (representative of urban and suburban scenarios) and consider different SNR levels ($\delta = 10, 13, 16, 19, 22, 25$ dB) under low-demand conditions. Figure 9 showcases the impact of different γ values on resource allocation, demonstrating the STRAWS method's ability to adjust to varying propagation conditions. The results show that as the path-loss exponent increases, the SAS may allocate more power to the eligible users to compensate for signal loss and maintain reliable communication links. If, during the trade-off between channel and power allocation, the SAS detects that more power is inefficient then the SAS rejects the user's request and does not allocate any channels to it. Thereby, SAS might reduce the number of allocated channels when path loss is higher. For example, it can be seen that in high SNR ($\delta = 25$ dB), changing path-loss exponent γ from 2 to a random parameter between $[2, 3]$ for every user causes nearly a 30% decrease in the number of channels allocated (from 35.45 to 25.36 on average). At the same time, while the total power utilized decreased from 2.71W to 2.01W, the power allocated to individual channels increased by 3.5%. Similarly, by increasing the path-loss exponent γ from 2 to 3, we can observe a 55% decrease (from 35.45 to 16.33) in the number of channels allocated and a nearly 40% increase (from 2.71 to 1.74) in allocated power to every individual user. These results once again indicate the system's ability to tailor dynamic resource allocation to specific propagation characteristics.

VI. Conclusion

In this paper, we developed a deterministic optimization approach for the coordinated utilization of spectrum bandwidth resources by allocating channels to users and managing their transmit power. The approach is developed for the CBRS setting where a central arbiter SAS manages the spectrum resources. We studied four configurations that differ in how the channels and transmit power are allocated. Specifically, we developed models for contiguous and non-contiguous approaches to channel allocation and uniform and nonuniform approaches to power allocation. In all our models, we ensure compliance with permissible interference levels among GAA users, minimum power sensitivity of CBSDs, and minimum SNR threshold values. Through extensive numerical experiments, we established that non-contiguous-channel and non-uniform-power allocation outperform other configurations. To address the computational challenges associated with solving optimization problems, we proposed the STRAWS method, which identifies a solution that can be used to warm-start the optimization solvers. Numerical experiments reveal that STRAWS enables MILP solvers to identify better solutions within the computational time limits. Our analyses provided additional insights regarding the trade-offs between total network transmit power and the number of transmission requests supported. This research holds the potential for various extensions, including applying the framework to domains with new spectrum and interference levels, such as cognitive radios, IoT networks, and other emerging wireless technologies, to enhance decision-making in the optimization process. In our future research, we will explore an optimization problem for resource allocation that incorporates Tapped-Delay Lines (TDL). This approach aims to enhance resource management in communication systems, addressing challenges posed by doubly-selective fading and time-varying channels, ultimately contributing to more adaptive and efficient wireless communication environments.

References

[1] *Notice of Proposed Rulemaking and Order in the Matter of Amendment of the Commission's Rules With Regard to Commercial Operations in the 3550-3650 MHZ Band*, Federal Commun. Comm., Washington, DC, USA, 2012.

[2] A. Sahoo, "Fair resource allocation in the citizens broadband radio service band," in *Proc. IEEE Int. Symp. Dyn. Spectr. Access Netw. (DySPAN)*, 2017, pp. 1–2.

[3] M. Chen, A. Liu, W. Liu, K. Ota, M. Dong, and N. N. Xiong, "RDRL: A recurrent deep reinforcement learning scheme for dynamic spectrum access in reconfigurable wireless networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 2, pp. 364–376, Mar./Apr. 2022.

[4] J. A. Stine and C. E. Caicedo Bastidas, "Enabling spectrum sharing via spectrum consumption models," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 4, pp. 725–735, Apr. 2015.

[5] H. Nam, K. Han, J. Park, H. Cha, and S.-L. Kim, "A spectrum sharing system: Visualized interference classification and CNN approach," in *Proc. IEEE Int. Symp. Dyn. Spectr. Access Netw. (DySPAN)*, 2021, pp. 33–40.

[6] S.-Y. Liu, H.-S. Lin, and C.-Y. Huang, "Design and implement domain proxy based CBRS system for 5G," in *Proc. IEEE VTS Asia Pac. Wireless Commun. Symp. (APWCS)*, 2019, pp. 1–6.

[7] K. S. Manosha et al., "A channel allocation algorithm for citizens broadband radio service/spectrum access system," in *Proc. Eur. Conf. Netw. Commun. (EuCNC)*, 2017, pp. 1–6.

[8] "Licensed shared access (LSA)," Eur. Conf. Postal Telecommun. Admin., Electron. Commun. Comm., Copenhagen, Denmark, Rep. ECC 205, 2014. [Online]. Available: http://www.erodocdb.dk/Docs/doc98/official/pdf/ECCREP205.PDF

[9] P. Agarwal, A. Kumar, and R. S. Yamaguchi, "Preserving operation frequency privacy of incumbents in CBRS," *IEEE Access*, vol. 10, pp. 111022–111041, 2022.

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.

[10] "Amendment of the commissions rules with regard to commercial operations in the 3550-3650 MHz band, report and order and second further notice of proposed rulemaking," Dept. Eng. Technol. Wireless Telecommun., Federal Commun. Comm., Washington, DC, USA, document 12-354, 2015.

[11] C. Tarver et al., "Enabling a use-or-share framework for PAL–GAA sharing in CBRS networks via reinforcement learning," *IEEE Trans. Cogn. Commun. Netw.*, vol. 5, no. 3, pp. 716–729, Sep. 2019.

[12] N. Jai et al., "Optimal channel allocation in the CBRS band with Shipborne radar incumbents," in *Proc. IEEE Int. Symp. Dynamic Spectr. Access Netw. (DySPAN)*, 2021, pp. 80–88.

[13] M. Grissa, A. A. Yavuz, B. Hamdaoui, and C. Tirupathi, "Anonymous dynamic spectrum access and sharing mechanisms for the CBRS band," *IEEE Access*, vol. 9, pp. 33860–33879, 2021.

[14] H. Hu and R. Chai, "Blockchain-based hierarchical spectrum sharing architecture and resource allocation algorithm for CBRS system," in *Proc. IEEE 94th Veh. Technol. Conf. (VTC)*, 2021, pp. 1–5.

[15] W. Abbass, R. Hussain, J. Frnda, I. L. Khan, M. A. Javed, and S. A. Malik, "Optimal resource allocation for GAA users in spectrum access system using Q-learning algorithm," *IEEE Access*, vol. 10, pp. 60790–60804, 2022.

[16] H. Hui, X. Guo, H. Wang, and W. Chen, "Spectrum sharing in cache-empowered wireless networks: Reservation-based versus auction-based approaches," *IEEE Trans. Cogn. Commun. Netw.*, vol. 9, no. 5, pp. 1126–1140, Oct. 2023.

[17] L. Xiao, Y. Li, C. Dai, H. Dai, and H. V. Poor, "Reinforcement learning-based NOMA power allocation in the presence of smart jamming," *IEEE Trans. Veh. Technol.*, vol. 67, no. 4, pp. 3377–3389, Apr. 2018.

[18] Ł. Kułacz et al., "Coordinated spectrum allocation and coexistence management in CBRS-SAS wireless networks," *IEEE Access*, vol. 7, pp. 139294–139316, 2019.

[19] Y. Wang, X. Ding, and G. Zhang, "A novel dynamic spectrum-sharing method for GEO and LEO satellite networks," *IEEE Access*, vol. 8, pp. 147895–147906, 2020.

[20] W. Qi, Q. Song, L. Guo, and A. Jamalipour, "Energy-efficient resource allocation for UAV-assisted vehicular networks with spectrum sharing," *IEEE Trans. Veh. Technol.*, vol. 71, no. 7, pp. 7691–7702, Jul. 2022.

[21] M. He, J. Xu, W. Xu, H. Shen, N. Wang, and C. Zhao, "RIS-assisted quasi-static broad coverage for wideband mmWave massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 22, no. 4, pp. 2551–2565, Apr. 2023.

[22] Z. Youssef et al., "Concept design of medium access control for spectrum access systems in 3.5 GHz," in *Proc. Int. Conf. Wireless Commun., Signal Process. Netw. (WiSPNET)*, 2018, pp. 1–8.

[23] M. Rahman, M. Yuksel, and T. Quint, "A game-theoretic framework to regulate freeriding in inter-provider spectrum sharing," *IEEE Trans. Wireless Commun.*, vol. 20, no. 6, pp. 3941–3957, Jun. 2021.

[24] S. Zheng, T. Han, Y. Jiang, and X. Ge, "Smart contract-based spectrum sharing transactions for multi-operators wireless communication networks," *IEEE Access*, vol. 8, pp. 88547–88557, 2020.

[25] M. Patnaik, G. Prabhu, C. Rebeiro, V. Matyas, and K. Veezhinathan, "ProBLeSS: A proactive blockchain based spectrum sharing protocol against SSDF attacks in cognitive radio IoBT networks," *IEEE Netw. Lett.*, vol. 2, no. 2, pp. 67–70, Jun. 2020.

[26] P. Kaurav, S. K. Koul, and A. Basu, "Determination of near field path loss in millimeter wave spectrum," in *Proc. IEEE MTT-S Int. Microw. RF Conf. (IMARC)*, 2019, pp. 1–4.

[27] L. L. Wang, "A mitigation technique for adjacent channel interference in communication satellites," in *Proc. 36th Int. Commun. Satell. Syst. Conf. (ICSSC)*, 2018, pp. 1–4.

[28] G. P. McCormick, "Computability of global solutions to factorable nonconvex problems: Part I: Convex underestimating problems," *Math. Program.*, vol. 10, pp. 147–175, Dec. 1976.

[29] G. Nemhauser and L. Wolsey, "Computational complexity," in *Integer and Combinatorial Optimization*. Hoboken, NJ, USA: Wiley, 1988, pp. 114–145, ch. I.5,II. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118627372.ch5

[30] (U.S. Gov. Publ. Office, Washington, DC, USA). *Code of Federal Regulations—Title 47, Part 96, Subpart E, Section 96.41*. (Dec. 7, 2018). [Online]. Available: https://www.ecfr.gov/current/title-47/chapter-I/subchapter-D/part-96/subpart-E/section-96.41

[31] "Mesh rider radio platform CBRS 3550–3700 MHz datasheet," Data Sheet, Los Angeles, CA, USA, 2023. [Online]. Available: https://doodlelabs.com/datasheets/mesh-rider-radio-platform-CBRS-3550-3700-MHz-datasheet/



Zahra Hoobakht received the B.S. degree in electrical engineering from the Iran University of Science and Technology, Tehran, Iran, and the M.S. degree in electrical engineering from the Amirkabir University of Tehran (Tehran Polytechnic), Tehran. She is currently pursuing the Ph.D. degree in electrical engineering with Southern Methodist University, Dallas, TX, USA. She investigates the application of optimization techniques and machine learning for 5G wireless communication and beyond. Her research interests include resource allocation for wireless networks, mobile-edge computing networks, data science, and computer vision.



Harsha Gangammanavar received the M.S. degree in electrical engineering and the Ph.D. degree in operations research from The Ohio State University, Columbus, Ohio. He is an Associate Professor of Operations Research and Engineering Management with Southern Methodist University (SMU), Dallas, TX, USA. He is an Affiliate Faculty with SMU's Data Science Institute and was a Visiting Assistant Professor with the University of Southern California from 2013 to 2015. He has developed algorithms for addressing planning and operations problems in infrastructure, healthcare, and communication systems. His research interests are in computational stochastic programming and large-scale optimization.



Dinesh Rajan (Senior Member, IEEE) received the B.Tech. degree in electrical engineering from the Indian Institute of Technology Madras, Chennai, and the M.S. and Ph.D. degrees in electrical and computer engineering from Rice University, Houston, TX, USA. In 2002, he joined Southern Methodist University (SMU) as an Assistant Professor. He served as the Department Chair with the Department of Electrical and Computer Engineering, SMU from 2012 to 2022, where he is currently an Associate Dean and the Cecil and Ida Green Professor with the Lyle School of Engineering. His broad interests are in the sensing, extraction, transmission, and dissemination of information. His current research is interdisciplinary and spans the traditional areas of communications theory, wireless networks, information theory, system optimization, and computational imaging.

Authorized licensed use limited to: Southern Methodist University. Downloaded on November 04,2026 at 00:40:41 UTC from IEEE Xplore. Restrictions apply.